# LAND & LEDGERS

# HISTORICAL WORK, CONSTRUCTION & INFRASTRUCTURE CALIBRATION REFERENCE

Dakota Territory / South Dakota and comparable North American practice, 1870-1905

Supplemental historical reference for Codex and implementation calibration

**STATUS - SUPPLEMENTAL REFERENCE, NOT DESIGN AUTHORITY**

This document does not supersede or amend the Canonical Project Bible or the Technical Implementation Bible. Canon XXVIII controls intended gameplay. Technical XI controls implementation architecture. This reference exists only to preserve historical timing, productivity and physical-work evidence so implementation can choose plausible calibration ranges.

Revision date: October 5, 2026

Research window: 1870-1905, with a small number of clearly labeled pre-1870 precedents used only where they illuminate period practice.

Primary emphasis: Dakota Territory / South Dakota / North Dakota, supplemented by period U.S. engineering manuals and comparable frontier railroad practice.

## Core rule

Use this document to answer questions such as “how long should this take?” by finding the closest physical analogue, then adjusting for actual scope, crew, methods, tools, material availability, freight, weather, terrain, finance and interruption. Never convert one historical example or record-setting performance into a universal constant.

# PART I - PURPOSE, METHOD & HOW CODEX SHOULD USE THIS

## 1.1 What this reference is for

Land & Ledgers already defines construction and infrastructure as physical work performed through real people, materials, equipment, logistics, agreements and projects. This reference adds historical orders of magnitude. Its purpose is calibration: to stop implementation from inventing implausible durations or treating an exceptional record as an ordinary work rate.

## 1.2 Active work is not elapsed project time

Every timing estimate should distinguish four clocks whenever the evidence permits:

- Active work: time in which crews are actually performing the relevant task or work package.
- Elapsed project time: first meaningful work through usable or accepted completion, including idle periods.
- Lead and queue time: waiting for materials, transport, specialists, approvals, financing or a free crew.
- Interruption time: weather, freeze, flood, breakdown, dispute, rework or other causal stoppage.

**CODEX RULE**
A historical statement that “a railroad laid three miles in a day” normally describes track laying on prepared grade with ties, rail and logistics already staged. It does not mean three miles of complete operating railway were created from untouched prairie in one day.

## 1.3 Evidence classes

| Class | Meaning | Use |
|---|---|---|
| A - direct dated record | Court record, government engineering record, period manual or dated contemporary account with concrete quantities/times. | May anchor a narrow range directly. |
| B - strong regional history | State/NPS historic-property research, local historical atlas or first-person memoir preserving useful dates and scope. | Good regional calibration, but inspect scope carefully. |
| C - comparable period practice | Period North American evidence outside Dakota or just outside 1870-1905. | Use to bound methods or throughput, never as a Dakota constant. |
| D - synthesis range | A recommended implementation envelope derived from multiple anchors. | Calibration guidance only. Recompute when better evidence appears. |

## 1.4 Default calibration policy

- Default to ordinary or moderately efficient conditions, not record performance.
- Treat materials already on site, prepared grade and uninterrupted funding as favorable assumptions that must be stated.
- Prefer task/unit rates when project scope varies widely. Earthwork should use cubic yards, track laying miles on prepared grade, roofing squares or shingles and framing/material quantity rather than one “construction speed.”
- Keep uncertainty visible. A range is better than fake precision when crew size, dimensions or interruptions are unknown.

# PART II - QUICK CALIBRATION TABLE

The following envelopes are recommended synthesis ranges for implementation and testing. They are not direct quotations and are not Canon locks. “Elapsed” assumes an ordinary project with reasonable material access unless the row says otherwise.

| Project / task | Scope assumption | Active rate | Elapsed range | Anchors | Confidence |
|---|---|---|---|---|---|
| Very small claim shanty / rough cabin | ~80-250 sq ft, simple frame/board shelter | 2-10 working days | 2 days to 2 weeks | B1, B2 | Medium |
| Simple small frame house | roughly 3-room frontier house | 15-35 working days | 3-6 weeks | B3 | Medium-high |
| Two-story frame storefront / store-dwelling | modest Main Street wood frame, no major masonry plant | - | 8-14 weeks | B4 | Medium |
| Substantial finished dwelling | contractor-supplied labor/materials, multiple finish stages | - | 16-24 weeks | B5 | High for elapsed order |
| 40 x 60 plank-framed barn | 5-person crew, material staged | 8-15 working days | 2-4 weeks | B6 | High |
| Small brick institutional/commercial building | brick walls, finished interior, modest tower/detail | - | 14-24 weeks | B7 | Medium-high |
| Funding-constrained small church | 44 x 30 ft example | - | ~9 months observed | B8 | High for example |
| Rail track laying on prepared grade | ties/rails staged, organized gang | 0.5-1.5 mi/day ordinary | same-day throughput | R1-R4 | High |
| Aggressive track laying on prepared grade | large organized gang, supply train keeps pace | 2-3+ mi/day | same-day throughput | R2, R5 | High as upper band |
| Manual earth loading | average laborer, 10-hour day | 15-25 yd3/man-day | task rate | R6 | High |
| Wheel scraper earth haul | 200-300 ft haul | 45-50 yd3/team-day | task rate | R7 | High |
| Large steam shovel loading | haul system keeps up | up to ~3,000 yd3/10 hr | task rate | R6 | High upper capacity |
| Major Missouri River railroad bridge | deep-water permanent truss crossing | - | ~3 years observed | R8 | High |
| Transcontinental telegraph open-route construction | survey/poles/wire coordinated | 3-8 mi/day | same-day route progress | U1 | High but pre-1870 precedent |

**IMPORTANT**

For a two-story storefront, use roughly 8-14 weeks as a starting elapsed envelope only when the design resembles a modest wood-frame frontier store and materials are reasonably accessible. For a small crude claim shanty, days can be historically defensible. Those two project classes should not share one building-time formula.

# PART III - BUILDINGS: SMALL SHELTER TO STORE

## 3.1 Claim shanty / very small rough cabin

A South Dakota pioneer recollection describes building a claim shanty in early May 1882 and states that it required about two days of work. The same account separately describes a more substantial 16 x 24 ft, 10-ft-studded house whose construction was inconvenienced by cold, wet spring weather. [B1]

South Dakota museum sources confirm how small these shelters could be: the Mydland claim shanty was 8 x 10 ft and another 1882 claim-house example was 16 x 10 ft. Such structures were intentionally crude and could be made from hauled lumber with minimal finish. [B2]

| Calibration case | Recommended starting envelope | Assumptions |
|---|---|---|
| Bare claim shanty / 8 x 10 to ~12 x 12 | 2-5 active working days | Materials at site, one or two capable builders/helpers, minimal floor/finish, no cellar or substantial masonry. |
| Small rough cabin / 150-250 sq ft | 4-10 active working days | Simple frame or board construction, basic door/window and roof, limited finish. |
| Weather/material constrained | 1-3+ elapsed weeks | Same physical work, but hauling, rain/cold, missing lumber or solo labor stretches calendar time. |

**Do not overgeneralize**
A two-day claim shanty is evidence that minimal shelter can be very fast. It is not evidence that a finished year-round cottage with floor, chimney, partitions, insulation and proper windows should also take two days.

## 3.2 Simple frame house

A first-person account of North Dakota settlement in 1889 reports that after a cellar was dug the household hired a carpenter to help. The writer states, “Our house was built in twenty-two days,” describing a simple frame house with one large room and two smaller rooms. [B3]

For implementation, a simple small frame house with materials and financing already available should therefore live in a weeks-scale envelope, not a season-scale default. A 15-35 working-day active envelope and roughly 3-6 weeks elapsed is a reasonable synthesis when scope is comparable.

# PART IV - BUILDINGS: STOREFRONTS, DWELLINGS & INSTITUTIONS

## 4.1 Two-story frontier frame storefront / store-dwelling

Charles Ingalls' second De Smet town building is a particularly useful regional anchor. Reconstruction research based on documents and newspapers dates construction from May to July 20, 1880. The building was a two-story frame structure used as a store-type town building and residence. [B4]

That is about 80 calendar days from May 1 to July 20, or roughly 11.4 weeks if "May" is conservatively treated as May 1. Because the exact May start date is not secure, the historical evidence supports an order of magnitude rather than an exact duration.

| Implementation interpretation | Range |
|---|---|
| Favorable modest two-story frame storefront | ~6-10 weeks elapsed if plans are simple, lumber/components are on hand and specialist work is light. |
| Ordinary reference envelope | ~8-14 weeks elapsed. |
| More finished / larger / delayed | ~12-20+ weeks if foundation, elaborate frontage, masonry, specialized glazing/heating, poor supply or weather adds burden. |

The range above is a synthesis. Codex should still decompose the actual storefront into site/foundation, frame, enclosure, roof, openings, stairs, interior finish and business-specific fixtures.

## 4.2 Substantial finished dwelling

In McCormack v. Phillips, an April 3, 1882 Sioux Falls contract required builder Samuel McCormack to furnish materials and construct a dwelling ready for occupation by September 1, subject to necessary weather interruption. The base contract was $4,770. The dispute later recorded more than $1,000 in extras/changes plus a separate barn/outbuilding contract. [B5]

The contractual window was 151 days, about 21.6 weeks. This is much longer than the 22-day simple frame-house example because "house" does not define one scope. Size, finish, supplied materials, multiple trades and owner changes matter.

**Recommended substantial-dwelling default**

Use roughly 16-24 weeks elapsed as a starting envelope for a genuinely finished, contractor-delivered dwelling of materially greater scope than a simple claim house. Adjust with actual quantities and staged dependencies.

# PART V - BUILDINGS: BARNS & INSTITUTIONAL WORK

## 5.1 40 x 60 plank-framed barn

The South Dakota State Historic Preservation Office farm-barn study records general agreement among barn experts that a journeyman carpenter with a crew of four men could raise a 40 x 60 ft plank-framed barn in about one week, with roughly another week for finishing. [B6]

This is one of the strongest construction-productivity anchors in the reference because it provides both structure size and crew form. The evidence is best used after foundation/site work and materials are ready.

| Stage | Calibration |
|---|---|
| Frame raising / major shell | ~1 working week with 1 journeyman + 4 men under favorable prepared conditions. |
| Finishing | ~1 additional working week. |
| Recommended active envelope | 8-15 workdays for comparable plank-frame barn shell/finish. |
| Recommended elapsed envelope | 2-4 weeks once site and materials are ready; longer when foundation, hauling, weather or procurement are included. |

## 5.2 Brick church / institutional building

The First Congregational Church of Milbank, a brick Gothic Revival structure, began construction in May 1883 and was completed by September. [B7] That supports roughly a four-month order of magnitude for a well-supported modest brick institutional project.

## 5.3 Funding delay can dominate elapsed time

A Huron church measuring 44 x 30 ft was begun October 19, 1881 but, explicitly “for lack of funds,” was not completed until July 30, 1882. [B8] That is about 284 days, or 40.6 weeks. The physical building did not suddenly require nine months of continuous labor; the financing interruption became project time.

**Calibration consequence**

Project duration should never be derived only from labor-hours. A well-funded four-month brick project and a smaller nine-month funding-constrained church can both be historically plausible.

# PART VI - COMPONENT PRODUCTIVITY & CREW EFFECTS

## 6.1 Roofing shingles

The March 1880 trade journal Carpentry and Building discussed shingling output directly. One correspondent treated claims of 3,000-4,000 shingles per day skeptically and wrote that laying about 2,000 shingles in a ten-hour day required a quick, active man. Another 1880 discussion reported roughly 2,500 shingles/day on a large straight roof for men assigned to shingling. [C1]

An 1884 estimating discussion again used roughly 2,000 shingles/day for straight work and about five roof/floor squares of certain simple work as estimating guidance. [C2]

| Use | Recommended calibration |
|---|---|
| Ordinary straight wood-shingle work | ~1,800-2,500 shingles per capable worker-day as a starting band. |
| Exceptional claims | 3,000-4,000+ may exist, but should not be default. |
| Complex roof / cuts / valleys / poor material | Reduce rate materially; use geometry and task difficulty rather than a flat roof-speed modifier. |

## 6.2 Crew composition

Period sources repeatedly show that skilled and helper labor are not interchangeable. The barn anchor uses one journeyman directing four men. A crew can therefore gain output when helpers handle staging, lifting and repetitive work, but specialist tasks remain limited by skilled capacity. Additional labor eventually encounters congestion and limited work fronts.

## 6.3 Bottom-up duration

For Codex, building duration should be assembled from meaningful quantities and work packages. A roofing package can use roof area and shingle coverage. Earthwork can use cubic yards and haul distance. Framing can use structure/material quantity plus crew capability. Openings and finish can use count/area. Overall elapsed time then comes from dependencies and availability, not from one building-type timer.

# PART VII - RAILWAY CONSTRUCTION: DO NOT CONFUSE TRACK LAYING WITH RAILROAD BUILDING

## 7.1 Railway work is a pipeline

| # | Work family | What it means |
|---|---|---|
| 1 | Survey and alignment | Reconnaissance, line, grades, right-of-way/authority and staking. |
| 2 | Clearing and preparation | Vegetation, access and work roads where required. |
| 3 | Grading / earthwork | Cuts, fills, embankments and subgrade. |
| 4 | Drainage and structures | Culverts, trestles, bridges and other crossings. |
| 5 | Track material staging | Ties, rails, fastenings and supply trains to end-of-track. |
| 6 | Track laying | Ties/rails positioned, joined and spiked on prepared grade. |
| 7 | Follow-up surfacing | Additional ties where method uses skeleton track, alignment, leveling, ballast and tamping. |
| 8 | Operating support | Stations, water/fuel, telegraph/signals and inspection where required. |

**CODEX LOCK FOR THIS REFERENCE**
A track-laying rate applies only to phase 6 unless the source explicitly says otherwise. Never multiply “miles per day of track laying” by calendar days and call the result completed railway through untouched terrain.

## 7.2 Physical material intensity

Period railroad specifications and government reports commonly show about 2,500 ties per mile. A Union Pacific government report recorded 2,640 ties/mile on a surveyed section and described 50-56 lb/yard iron rail. [R2] George Vose’s railroad-construction handbook likewise states about 2,500 ties per mile. [R3]

This physical material requirement is important: high track-laying rates are possible only when thousands of ties, hundreds of rails and fastenings are already delivered forward at the rate the gang consumes them.

# PART VIII - RAILWAY TRACK-LAYING CALIBRATION

## 8.1 Ordinary prepared-grade benchmark

Contemporary newspapers give a very useful ordinary benchmark. In March 1900, reports on a 25-mile Texas railroad extension said the remainder was about ready for ties and rails and track was being put down at one mile per day. [R1] An 1884 report on the Toledo, Ann Arbor and Northern Michigan likewise said track laying would be pushed at a mile per day. [R4]

## 8.2 Organized high-output gangs

A U.S. government report on the Union Pacific described track laying as often three miles per day, with a season average of 50 miles per month, attributing the speed to a large force working in a highly organized system. [R2] A late-1880s account of track laying from Great Falls toward Helena reported 98 miles in about a month of working days, roughly 3.25 miles per working day, with individual days above seven miles. [R5]

| Band | Prepared-grade track-laying rate | When to use |
|---|---|---|
| Conservative / disrupted | 0.25-0.75 mi/day | Material gaps, small gang, curves/crossings, interruptions or weak logistics. |
| Ordinary default | 0.75-1.5 mi/day | Prepared grade, ordinary organized gang and reliable forward supply. |
| Strong organized gang | 1.5-3 mi/day | Large practiced gang, long simple alignment and strong supply train. |
| Exceptional / record | 3-8+ mi/day | Special favorable runs or competitive record attempts. Never default. |

## 8.3 Calendar planning rule

For a 10-mile branch on already prepared prairie grade, one mile/day does not imply “10 days from untouched land to operation.” It implies about ten working days for the track-laying phase itself. Survey, grading, drainage, structures, procurement, material staging, surfacing and commissioning add their own calendars.

# PART IX - RAILWAY & ROAD EARTHWORK

## 9.1 Manual loading

Walter Loring Webb's 1900 Railroad Construction reviewed several authorities on hand loading and accepted an average of roughly 15-25 cubic yards of earth loaded into carts by an average laborer in a ten-hour day. [R6]

## 9.2 Horse-drawn wheel scrapers

Ira O. Baker's 1903 Treatise on Roads and Pavements gives approximately 50 cubic yards/day per wheel scraper at a 200-ft haul and about 45 cubic yards/day per team at a 300-ft haul. [R7]

## 9.3 Steam shovel

Webb gives about 3,000 cubic yards in ten hours as the capacity of the larger steam shovels, while explicitly noting that the haul system must remove material as fast as the shovel loads it. [R6] Treat 3,000 yd3/day as plant capacity under favorable support, not a universal achieved average.

| Method | Starting unit rate | Primary constraints |
|---|---|---|
| Hand loading to carts | 15-25 yd3 / laborer / 10 hr | Soil, management, cart supply, fatigue and haul cycle. |
| Wheel scraper, 200 ft haul | ~50 yd3 / scraper-day | Team count, loading/holding arrangement, grade and soil. |
| Wheel scraper, 300 ft haul | ~45 yd3 / team-day | Longer cycle and haul conditions. |
| Large steam shovel | up to ~3,000 yd3 / 10 hr | Continuous cars/haul, fuel/water, crew, cut geometry and material. |

## 9.4 Worked geometry example - not a historical "typical mile"

Suppose a deliberately simplified one-mile earth-road or railway formation required an average one-foot net cut/fill over a 24-ft width. Geometry alone gives about 4,693 cubic yards (24 x 5,280 x 1 / 27). Eight scraper teams averaging 47.5 yd3/day would move about 380 yd3/day, implying about 12.4 working days of scraper production before allowing for drainage, compaction, setup, difficult material, crossings or idle time.

**Why this matters**
Codex should calculate grading time from actual earth quantity and method. "Railroad grading = X days per mile" is only safe if the authored terrain/cross-section already implies the same earthwork volume.

# PART X - RAILWAY BRIDGES, TRESTLES & MAJOR STRUCTURES

## 10.1 Bismarck Missouri River Bridge

The Library of Congress HAER record describes the Northern Pacific Bismarck Bridge as the first bridge across the Missouri River in Dakota Territory. Construction was initiated in 1880 and took approximately three years to complete. The crossing used three 400-ft Whipple trusses on four massive granite piers and exceeded 1,500 ft overall. [R8]

This is an anchor for a major deep-water permanent railroad crossing, not a small creek bridge. A 1,500-ft Missouri crossing belongs in a years-scale project class because foundations, caissons/piers, fabricated iron/steel, erection and river logistics dominate.

| Structure class | Recommended implementation order of magnitude |
|---|---|
| Small timber culvert / very short bridge | Days to weeks once timber and crew are available; derive from actual span and foundation work. |
| Ordinary timber trestle / moderate crossing | Weeks to months; use pile/bent count, approach grading and material supply. |
| Large permanent river bridge | Many months to several years. Major Missouri-class crossings can be ~2-3 years. |
| Mega-project trestle/causeway | Project-specific. The 1902-1904 Ogden-Lucin Cutoff used ~3,000 men and a nearly 12-mile trestle within a two-year system build, demonstrating what massive continuous resources can accomplish. [R9] |

## 10.2 Resource concentration changes calendar time

The Ogden-Lucin Cutoff is useful precisely because it is not ordinary. Southern Pacific concentrated thousands of workers, specialized pile drivers, locomotives, dump cars and continuous operations on a single project. [R9] It should establish what is possible at extreme organizational scale, not the default productivity of a frontier contractor.

# PART XI - ROADS, TELEGRAPH & LINEAR INFRASTRUCTURE

## 11.1 Earth roads

Road construction should use the same unit-based earthwork approach as railway grading. Baker's 1903 road manual provides explicit scraper productivity by haul distance, allowing the game to derive time from road width, cut/fill, drainage and material rather than assigning a universal road-mile timer. [R7]

| Road condition | Calibration approach |
|---|---|
| Trail / cleared wagon path | Vegetation clearing, obstacle removal, limited earth shaping and crossings. Can advance quickly over easy open prairie. |
| Graded earth road | Compute excavation/embankment volume, ditching, culverts and scraper/crew throughput. |
| Improved gravel/macadam road | Add quarry/source, hauling, spreading, rolling and drainage. Surface material logistics can dominate. |

## 11.2 Telegraph line

The National Park Service account of the 1861 transcontinental telegraph reports that coordinated crews surveyed/marked the route, dug holes, installed poles and strung wire at roughly 3-8 miles of line per day. [U1] This is a pre-1870 precedent but close enough technologically to be useful as an upper/open-country reference for the game period.

| Telegraph context | Recommended starting band |
|---|---|
| Open prairie, poles/material staged, organized crew | ~3-6 route miles/day. |
| Highly favorable / large coordinated operation | Up to ~8 mi/day supported by the 1861 precedent. |
| Timber, broken terrain, crossings, poor pole supply | Substantially below 3 mi/day; separate clearing/haul bottlenecks. |
| Railroad-following telegraph | Can advance with the railway if poles/wire and a dedicated crew are continuously supplied; do not treat it as free. |

**Linear-infrastructure rule**
For roads, rail and telegraph, the slowest sustained upstream stage usually governs route completion. Fast laying/stringing crews cannot outrun grading, bridges, pole supply or material delivery indefinitely.

# PART XII - WORKED CODEX CALIBRATION EXAMPLES

## 12.1 Small claim cabin

| Input | Example |
|---|---|
| Scope | 10 x 14 ft rough frame/board claim shelter, one door, one small window, minimal finish. |
| People | One capable builder plus one helper. |
| Materials | Lumber, nails and roofing already delivered. |
| Historical anchor | Two-day South Dakota claim shanty [B1], plus 8 x 10 and 16 x 10 regional shanty forms [B2]. |
| Suggested implementation forecast | 3-6 active days; 3-10 calendar days in favorable weather. Add hauling or missing components separately. |
| Do not include automatically | Cellar, chimney, plaster, insulation, interior partitions or furniture unless the authored building requires them. |

## 12.2 Two-story frame storefront

| Input | Example |
|---|---|
| Scope | Modest two-story wood-frame Main Street building with ground-floor shop and upper rooms. |
| Historical anchor | Ingalls De Smet building constructed May-July 20, 1880 [B4]. |
| Suggested baseline | ~8-14 weeks elapsed under ordinary material access. |
| Faster case | ~6-8 weeks if design is simple, foundation is light, manufactured openings are on hand and crews are uninterrupted. |
| Slower case | ~14-20+ weeks with basement/masonry, elaborate storefront, specialized systems, shortages, bad weather or financing interruption. |
| Implementation method | Build from site/foundation -> frame -> roof/enclosure -> openings/stairs -> finish -> fixtures. Overall progress is a read model over those packages. |

# PART XIII - WORKED CODEX EXAMPLES: BARN & RAILWAY

## 13.1 40 x 60 plank barn

| Input | Example |
|---|---|
| Crew | 1 journeyman carpenter + 4 men. |
| Site/material state | Foundation/site ready; framing material delivered. |
| Historical anchor | ~1 week raising + ~1 week finishing [B6]. |
| Suggested baseline | ~10 working days active; 2-3 weeks elapsed once ready. |
| Blockers | Foundation not ready, lumber delayed, rain/storm, missing crew/equipment, later stall/fixtures. |
| Scaling rule | Adding labor helps only while there are parallel work fronts and competent coordination. |

## 13.2 One mile of simple prairie railroad

This example deliberately separates phases. It is not a historical claim that every prairie mile has these quantities.

| Phase | Illustrative calculation / timing logic |
|---|---|
| Survey/rights | Project-specific; may precede field construction by days to months. |
| Earthwork | If the authored formation contains 4,700 yd3 of movable earth, 8 scraper teams at ~45-50 yd3/day imply roughly 12-13 ideal production days before inefficiency/structures. |
| Drainage/culverts | Separate tasks based on actual crossings. Can overlap grading where geography allows. |
| Tie/rail staging | About 2,500 ties/mile is a period order of magnitude [R2-R3]. Freight must place them where the gang can consume them. |
| Track laying | Use ~0.75-1.5 mi/day ordinary prepared-grade throughput, so the track-laying phase for one mile is about one working day. |
| Surfacing/ballast/inspection | Separate follow-up. Skeleton track may carry construction trains before final surfacing, but this is not the same as finished main-line quality. |

**Result**
A one-mile railway segment can require weeks of calendar work even though the visible rail-laying gang can place the rail in roughly one day. That distinction is the central reason this reference exists.

# PART XIV - IMPLEMENTATION CALIBRATION RULES

## 14.1 Choose the analogue, then modify it

| Modifier | Direction | Examples |
|---|---|---|
| Scope/quantity | Direct | Larger footprint, more stories, more earth, longer span, more finish. |
| Method/technology | Direct/indirect | Balloon/plank framing, hand excavation vs scraper vs steam shovel, pile-driver access. |
| Crew capability | Mostly productivity/quality | Skilled lead, specialist scarcity, novice rework, supervision. |
| Crew size | Diminishing returns | More workers help only if there are independent work fronts and enough tools/material. |
| Material access | Calendar and cost | Local lumber vs rail-delivered doors/windows/brick/iron; stockouts and lead time. |
| Freight/logistics | Calendar | Railhead, wagon haul, river crossing, end-of-track supply train. |
| Weather/season | Task-specific | Frozen ground, mortar/plaster constraints, storms, flood, heat and daylight. |
| Finance/payment | Calendar | Funding shortfall can stop a smaller project for months even when remaining labor is modest. |
| Change/rework | Adds real work | Owner changes, hidden condition, failed inspection or defective workmanship. |

## 14.2 Stochastic variation should remain causal

If implementation needs variability, draw uncertainty around identifiable causes: quantity uncertainty, productivity, material arrival, weather and coordination. Avoid a single hidden ±X% “construction RNG” that cannot explain itself.

## 14.3 Keep three reported times

- Planned active work: crew-days or machine-days expected under current assumptions.
- Forecast elapsed completion: current calendar estimate including known queues and dependencies.
- Actual elapsed completion: historical result after interruptions, changes and rework.

## 14.4 Default/fast/slow presets for Codex testing

| Preset | Interpretation |
|---|---|
| Fast | Good crew, materials staged, simple design, favorable weather, no funding interruption. Use near the fast end of synthesis ranges, not records. |
| Ordinary | Competent ordinary operation with some idle/coordination time. This should be the usual regression fixture. |
| Slow but plausible | Thin labor market, haul delays, weather or finance issues. Preserve causal blockers rather than multiplying all task times uniformly. |
| Exceptional | Historic record or extraordinary resource concentration. Use only in tests of maximum plausible throughput. |

# PART XV - HISTORICAL ANCHOR REGISTER: BUILDINGS

**[B1]** OSU eHistory, “South Dakota - 1882 / Dakota Days” pioneer recollection. Early May claim shanty required about two days of work; same account describes a 16 x 24 ft, 10-ft-studded house and weather interference. https://ehistory.osu.edu/exhibitions/letters/working_hileman/100

**[B2]** South Dakota State University Agricultural Heritage Museum, “South Dakota Claim Shanty,” plus W. H. Over Museum Mydland Claim Shanty. Regional examples include tiny board shanties and an 8 x 10 ft 1882 structure. https://www.sdstate.edu/south-dakota-agricultural-heritage-museum/south-dakota-claim-shanty ; https://whovermuseum.org/exhibit-mydland-claim-shanty/

**[B3]** “Fifteen Years in North Dakota, 1883-1898,” first-person memoir preserved by CARLI/North Park University. Reports a simple frame house built in twenty-two days after hiring a carpenter. https://collections.carli.illinois.edu/digital/api/collection/npu_sahq/id/4678/download

**[B4]** Rich Kurz / Nansie Cleaveland, “Pa Ingalls’ Store - How to Recreate a Building That No Longer Exists,” and PioneerGirl research. Documents a two-story De Smet frame building constructed May-July 20, 1880. https://richkurz.com/history/DeSmet/PaStore/Recreating_Pa_Store--RKurz.pdf ; https://www.pioneergirl.com/blog/archives/6190

**[B5]** McCormack v. Phillips, Dakota Territorial Supreme Court (1887), arising from an April 3, 1882 Sioux Falls construction agreement. Finished dwelling due September 1, base price $4,770, with documented extras/changes and separate outbuilding work. https://case-law.vlex.com/vid/mccormack-v-phillips-929887765

**[B6]** South Dakota State Historic Preservation Office, “Common Farm Barns of South Dakota” Multiple Property Documentation. Reports general agreement that a journeyman carpenter with four men could raise a 40 x 60 ft plank-framed barn in about one week with about another week for finishing. https://history.sd.gov/preservation/docs/FarmBarnsSouthDakotaMPS.pdf

**[B7]** Society of Architectural Historians, SAH Archipedia, First Congregational Church of Milbank. Brick Gothic Revival church construction began May 1883 and was completed by September. https://sah-archipedia.org/buildings/SD-01-051-0039

**[B8]** Andreas’ 1884 History of Beadle County, Huron church account. A 44 x 30 ft church begun October 19, 1881 was not completed until July 30, 1882 explicitly for lack of funds. https://beadlecountysd.blogspot.com/2011/02/andreas-history-of-beadle-county-1884_16.html

**[C1]** Carpentry and Building, March 1880. Period trade discussion of shingling: approximately 2,000 shingles/day for a quick active worker, with higher claims treated cautiously; another 1880 note reports 2,500/day on large straight roofs. https://www.usmodernist.org/CB/CB-1880.pdf

**[C2]** Carpentry and Building, March 1884. Period estimating discussion uses about 2,000 shingles/day for straight work and discusses task-specific labor estimation. https://upload.wikimedia.org/wikipedia/commons/3/35/Carpentry_and_building_%28IA_carpentrybuildin6188unse%29.pdf

# PART XVI - HISTORICAL ANCHOR REGISTER: RAIL, ROADS & UTILITIES

**[R1]** The Weekly Sentinel (Nacogdoches, Texas), March 15, 1900. A 25-mile extension reported track being put down at one mile/day with remaining work about ready for ties and rails. https://texashistory.unt.edu/ark:/67531/metapth1829760/m1/3/

**[R2]** U.S. Government report on the Union Pacific Railroad. Records 2,640 ties/mile on one section and says track laying was often three miles/day, averaging 50 miles/month through the season under a large, highly organized force. https://www.govinfo.gov/content/pkg/SERIALSET-01360_00_00-005-0004-0000/pdf/SERIALSET-01360_00_00-005-0004-0000.pdf

**[R3]** George L. Vose, Handbook of Railroad Construction. Period railroad specification states about 2,500 ties/mile and describes track-laying requirements. https://www.gutenberg.org/cache/epub/56688/pg56688-images.html

**[R4]** Ann Arbor Register, March 27, 1884. Reports track laying on the Toledo, Ann Arbor and Northern Michigan at a planned rate of one mile/day. https://aadl.org/files/documents/pdf/aa_register/aa_register_18840327.pdf

**[R5]** Charles Dudley Warner, Studies in the South and West, with Comments on Canada (1889), describing a late-1880s railroad push from Great Falls to Helena: 98 miles over the working period, about 3.25 miles/working day, with exceptional days over seven miles. https://www.arc-codex.com/library/52290

**[R6]** Walter Loring Webb, Railroad Construction: Theory and Practice (1900). Gives 15-25 yd3/10-hour day as a practical average range for hand loading earth and about 3,000 yd3/10 hours as capacity of larger steam shovels when haul keeps pace. Library of Congress: https://www.loc.gov/item/00002911/ ; digitized text: https://upload.wikimedia.org/wikipedia/commons/f/f0/Railroad_construction._Theory_and_practice_%28IA_railroadconstruc02webb%29.pdf

**[R7]** Ira O. Baker, A Treatise on Roads and Pavements (1903). Wheel scraper example: about 50 yd3/day at 200-ft haul, about 45 yd3/day at 300-ft haul. https://upload.wikimedia.org/wikipedia/commons/c/cf/A_treatise_on_roads_and_pavements_%28IA_treatiseonroadsp00bakerich%29.pdf

**[R8]** Library of Congress, Historic American Engineering Record, Bismarck Bridge. Initial construction 1880-1883; approximately three years; three 400-ft Whipple trusses on four massive granite piers. https://www.loc.gov/item/nd0023/

**[R9]** Library of Congress / HAER and SAH Archipedia, Ogden-Lucin Cutoff Trestle. Built 1902-1904; nearly 12-mile trestle; project used roughly 3,000 men and extraordinary logistics. https://www.loc.gov/pictures/item/ut0004/ ; https://sah-archipedia.org/buildings/UT-01-057-0074

**[U1]** U.S. National Park Service, “The Transcontinental Telegraph.” In 1861 coordinated crews installed about 3-8 miles/day. This is a clearly labeled pre-window technological precedent, not a Dakota 1870-1905 direct observation. https://home.nps.gov/articles/000/the-transcontinental-telegraph.htm

# PART XVII - LIMITATIONS, RESEARCH GAPS & FUTURE EXPANSION

## 17.1 What this edition can answer confidently

- Orders of magnitude for tiny claim shelters, simple frame houses, modest two-story frame town buildings, a substantial finished dwelling and a 40 x 60 plank-framed barn.
- Prepared-grade railroad track-laying throughput and the distinction between ordinary, aggressive and exceptional rates.
- Unit earthwork productivity for manual loading, horse-drawn wheel scrapers and large steam shovels around 1900.
- Years-scale duration of a major Missouri River railroad bridge and the resource intensity of a major 1902-1904 trestle/causeway project.
- Open-country telegraph construction order of magnitude from a closely comparable pre-1870 precedent.

## 17.2 Gaps that should remain explicit

| Gap | Current treatment |
|---|---|
| Exact two-story storefront labor-days by trade | Elapsed regional example exists, but crew size and detailed quantities remain incomplete. Use 8-14 weeks as synthesis, then decompose. |
| Small timber bridge/trestle rates | No strong Dakota unit-rate set yet. Derive from physical quantities and use broad weeks-to-months envelope. |
| Brick masonry output by mason-day in Dakota | Needs deeper period trade/manual research before locking a unit rate. |
| Sewer/water-main trenching feet/day | Known project/failure cases exist, but this edition does not yet claim a universal trenching rate. |
| Gravel/macadam miles/day | Use material volume, haul and rolling tasks rather than a guessed per-mile constant. |
| Winter construction productivity | Task-specific weather logic is better supported than one seasonal multiplier; exact modifiers remain calibration. |

## 17.3 Recommended future additions

If development reaches one of the gaps above, expand this reference with the same evidence discipline rather than creating another permanent document. Strong next targets are masonry output, excavation/foundation crews, small timber bridges/culverts, sewer/water trenching, commercial brick blocks and industrial machinery installation.

### FINAL USE RULE

This reference preserves historical evidence and recommended calibration envelopes. The simulation should still compute outcomes from actual world state. When a historical source and the authored physical situation differ, preserve the source as evidence and adjust the forecast through the real differences rather than forcing the world to match the source.

**END OF HISTORICAL WORK, CONSTRUCTION & INFRASTRUCTURE CALIBRATION REFERENCE**